

Unit 12. Kronecker Product, vec, and Patterned Matrices

1 The Kronecker Product

1.1 Definition and algebra

Definition 12.1 (Kronecker product). Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For $\mathbf{A} = (a_{ij}) \in \mathbb{F}^{m \times n}$ and $\mathbf{B} = (b_{rs}) \in \mathbb{F}^{p \times q}$, the **Kronecker product** is

$$\mathbf{A} \otimes \mathbf{B} := \begin{pmatrix} a_{11}\mathbf{B} & \cdots & a_{1n}\mathbf{B} \\ \vdots & & \vdots \\ a_{m1}\mathbf{B} & \cdots & a_{mn}\mathbf{B} \end{pmatrix} \in \mathbb{F}^{mp \times nq}.$$

Equivalently,

$$(\mathbf{A} \otimes \mathbf{B})_{p(i-1)+r, q(j-1)+s} = a_{ij}b_{rs}.$$

Origin: A&M §10.1; H §16.1; S §4.3.1; MC §2.3

Theorem 12.2 (Kronecker algebra). Let $\mathbb{F} \in \{\mathbb{R}, \mathbb{C}\}$. For $\alpha, \beta \in \mathbb{F}$ and conformable matrices over $\mathbb{F}$,

$$(\alpha \mathbf{A} + \beta \mathbf{C}) \otimes \mathbf{B} = \alpha(\mathbf{A} \otimes \mathbf{B}) + \beta(\mathbf{C} \otimes \mathbf{B}), \quad (1)$$

$$\mathbf{A} \otimes (\alpha \mathbf{B} + \beta \mathbf{D}) = \alpha(\mathbf{A} \otimes \mathbf{B}) + \beta(\mathbf{A} \otimes \mathbf{D}), \quad (2)$$

$$(\mathbf{A} \otimes \mathbf{B}) \otimes \mathbf{C} = \mathbf{A} \otimes (\mathbf{B} \otimes \mathbf{C}), \quad (3)$$

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = (\mathbf{AC}) \otimes (\mathbf{BD}), \quad (4)$$

$$(\mathbf{A} \otimes \mathbf{B})' = \mathbf{A}' \otimes \mathbf{B}'. \quad (5)$$

If $\mathbb{F} = \mathbb{C}$, then

$$(\mathbf{A} \otimes \mathbf{B})^* = \mathbf{A}^* \otimes \mathbf{B}^*. \quad (6)$$

Origin: Course formulation from A&M Exs. 10.3–10.4, 10.7; H Lemmas 16.1.1–16.1.2 and Eq. (16.1.15); S §4.3.1; MC Props. 3–4

Proof. Equations (1), (2), and (5) hold blockwise. For $\mathbf{A} \in \mathbb{F}^{m \times n}$, $\mathbf{B} \in \mathbb{F}^{p \times q}$, and $\mathbf{C} \in \mathbb{F}^{u \times v}$,

both sides of (3) have $u \times v$ block

$$a_{ij}b_{rs}\mathbf{C} \quad \text{in block position} \quad (p(i-1)+r, q(j-1)+s).$$

If $\mathbf{C} \in \mathbb{F}^{n \times u}$ and $\mathbf{D} \in \mathbb{F}^{q \times v}$, the (i, k) block of the left side of (4) is

$$\sum_{j=1}^n (a_{ij}\mathbf{B})(c_{jk}\mathbf{D}) = \left(\sum_{j=1}^n a_{ij}c_{jk} \right) \mathbf{BD} = (\mathbf{AC})_{ik}\mathbf{BD}.$$

For $\mathbb{F} = \mathbb{C}$, entrywise conjugation gives

$$\overline{\mathbf{A} \otimes \mathbf{B}} = \overline{\mathbf{A}} \otimes \overline{\mathbf{B}}.$$

Consequently,

$$(\mathbf{A} \otimes \mathbf{B})^* = \overline{\mathbf{A} \otimes \mathbf{B}}' = (\overline{\mathbf{A}} \otimes \overline{\mathbf{B}})' = \mathbf{A}^* \otimes \mathbf{B}^*.$$

□

Corollary 12.3 (Inverses, powers, and orthogonality). If $\mathbf{A} \in \mathbb{R}^{m \times m}$ and $\mathbf{B} \in \mathbb{R}^{n \times n}$ are nonsingular, then

$$(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}.$$

For $k \geq 0$,

$$(\mathbf{A} \otimes \mathbf{B})^k = \mathbf{A}^k \otimes \mathbf{B}^k.$$

If $\mathbf{A}$ and $\mathbf{B}$ are orthogonal, then $\mathbf{A} \otimes \mathbf{B}$ is orthogonal.

Origin: A&M Exs. 10.7–10.8; H Eqs. (16.1.22)–(16.1.23) and Ex. 16.4; MC Prop. 4

Proof. The mixed-product rule gives

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^{-1} \otimes \mathbf{B}^{-1}) = \mathbf{I}_m \otimes \mathbf{I}_n = \mathbf{I}_{mn},$$

$$(\mathbf{A} \otimes \mathbf{B})'(\mathbf{A} \otimes \mathbf{B}) = (\mathbf{A}'\mathbf{A}) \otimes (\mathbf{B}'\mathbf{B}) = \mathbf{I}_{mn}.$$

The power identity follows by induction; $k = 0$ uses $\mathbf{I}_m \otimes \mathbf{I}_n = \mathbf{I}_{mn}$. □

Example 12.1 (Two block orderings). For $\mathbf{A} = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$,

$$\mathbf{A} \otimes \mathbf{B} = \begin{pmatrix} 0 & 1 & 0 & 2 \\ 1 & 0 & 2 & 0 \\ 0 & 3 & 0 & 4 \\ 3 & 0 & 4 & 0 \end{pmatrix}, \quad \mathbf{B} \otimes \mathbf{A} = \begin{pmatrix} 0 & 0 & 1 & 2 \\ 0 & 0 & 3 & 4 \\ 1 & 2 & 0 & 0 \\ 3 & 4 & 0 & 0 \end{pmatrix}.$$

Thus $\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}$.

Origin: A&M Exs. 10.1–10.2; H §16.1

1.2 Singular values, spectrum, and order

Theorem 12.4 (Singular values and rank). Let the nonzero singular values of $\mathbf{A}$ and $\mathbf{B}$ be $\alpha_1, \dots, \alpha_r$ and $\beta_1, \dots, \beta_s$. The nonzero singular values of $\mathbf{A} \otimes \mathbf{B}$ are

$$\{\alpha_i \beta_j : 1 \leq i \leq r, 1 \leq j \leq s\}.$$

Consequently,

$$\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B}), \quad (7)$$

$$\|\mathbf{A} \otimes \mathbf{B}\|_F = \|\mathbf{A}\|_F \|\mathbf{B}\|_F, \quad (8)$$

$$\|\mathbf{A} \otimes \mathbf{B}\|_2 = \|\mathbf{A}\|_2 \|\mathbf{B}\|_2. \quad (9)$$

Origin: Course formulation from A&M Ex. 10.12(c); H Eq. (16.1.26) and Ex. 16.9

Proof. Take singular value decompositions

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}', \quad \mathbf{B} = \mathbf{P}\mathbf{\Gamma}\mathbf{Q}'.$$

Then

$$\mathbf{A} \otimes \mathbf{B} = (\mathbf{U} \otimes \mathbf{P})(\mathbf{\Sigma} \otimes \mathbf{\Gamma})(\mathbf{V} \otimes \mathbf{Q})'.$$

The outer factors are orthogonal. The nonzero diagonal entries of $\mathbf{\Sigma} \otimes \mathbf{\Gamma}$ are $\alpha_i \beta_j$; row and column permutations put them in decreasing order. Hence (7). Moreover,

$$\begin{aligned}\|\mathbf{A} \otimes \mathbf{B}\|_F^2 &= \sum_{i,j} \alpha_i^2 \beta_j^2 = \left(\sum_i \alpha_i^2 \right) \left(\sum_j \beta_j^2 \right), \\ \|\mathbf{A} \otimes \mathbf{B}\|_2 &= \max_{i,j} \alpha_i \beta_j = \left(\max_i \alpha_i \right) \left(\max_j \beta_j \right).\end{aligned}$$

□

Theorem 12.5 (Spectrum, trace, and determinant). Let $\mathbf{A} \in \mathbb{R}^{m \times m}$ have eigenvalues $\lambda_1, \dots, \lambda_m$ and let $\mathbf{B} \in \mathbb{R}^{n \times n}$ have eigenvalues $\mu_1, \dots, \mu_n$, counted with algebraic multiplicity over $\mathbb{C}$. Then $\mathbf{A} \otimes \mathbf{B}$ has eigenvalues

$$\lambda_i \mu_j, \quad 1 \leq i \leq m, \quad 1 \leq j \leq n.$$

Also,

$$\operatorname{tr}(\mathbf{A} \otimes \mathbf{B}) = (\operatorname{tr} \mathbf{A})(\operatorname{tr} \mathbf{B}), \quad (10)$$

$$\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^n (\det \mathbf{B})^m. \quad (11)$$

Origin: A&M Exs. 10.7, 10.10, 10.12(b); H Eqs. (16.1.25) and (16.3.18); S Ex. 4.2

Proof. Regard $\mathbf{A}$ and $\mathbf{B}$ as complex matrices. Complex Schur factorisations give unitary $\mathbf{Q}, \mathbf{R}$ and upper triangular $\mathbf{T}, \mathbf{S}$ such that

$$\mathbf{Q}^* \mathbf{A} \mathbf{Q} = \mathbf{T}, \quad \mathbf{R}^* \mathbf{B} \mathbf{R} = \mathbf{S}.$$

Thus

$$\begin{aligned}(\mathbf{Q} \otimes \mathbf{R})^*(\mathbf{A} \otimes \mathbf{B})(\mathbf{Q} \otimes \mathbf{R}) &= (\mathbf{Q}^* \otimes \mathbf{R}^*)(\mathbf{A} \otimes \mathbf{B})(\mathbf{Q} \otimes \mathbf{R}) \\ &= (\mathbf{Q}^* \mathbf{A} \mathbf{Q}) \otimes (\mathbf{R}^* \mathbf{B} \mathbf{R}) = \mathbf{T} \otimes \mathbf{S}.\end{aligned}$$

The right side is upper triangular with diagonal entries $\lambda_i \mu_j$. Equation (11) is their product. Directly from the diagonal blocks,

$$\mathrm{tr}(\mathbf{A} \otimes \mathbf{B}) = \sum_{i=1}^m \mathrm{tr}(a_{ii} \mathbf{B}) = (\mathrm{tr} \mathbf{A})(\mathrm{tr} \mathbf{B}).$$

□

Theorem 12.6 (Definiteness of a Kronecker product). Let $\mathbf{A}$ and $\mathbf{B}$ be real symmetric matrices.

- (i) If $\mathbf{A} \succeq \mathbf{O}$ and $\mathbf{B} \succeq \mathbf{O}$, then $\mathbf{A} \otimes \mathbf{B} \succeq \mathbf{O}$.
- (ii) If $\mathbf{A} \succ \mathbf{O}$ and $\mathbf{B} \succ \mathbf{O}$, then $\mathbf{A} \otimes \mathbf{B} \succ \mathbf{O}$.
- (iii) The same conclusions hold when both factors are respectively negative semidefinite or negative definite.
- (iv) Opposite definite signs give a negative definite product.

Origin: A&M Ex. 10.12(a); H Ex. 16.7

Proof. Choose orthonormal eigenbases $\mathbf{A}\mathbf{q}_i = \lambda_i\mathbf{q}_i$ and $\mathbf{B}\mathbf{u}_j = \mu_j\mathbf{u}_j$. Then

$$(\mathbf{A} \otimes \mathbf{B})(\mathbf{q}_i \otimes \mathbf{u}_j) = \lambda_i\mu_j(\mathbf{q}_i \otimes \mathbf{u}_j).$$

The mn vectors $\mathbf{q}_i \otimes \mathbf{u}_j$ are orthonormal because

$$(\mathbf{q}_i \otimes \mathbf{u}_j)'(\mathbf{q}_k \otimes \mathbf{u}_l) = (\mathbf{q}_i'\mathbf{q}_k)(\mathbf{u}_j'\mathbf{u}_l).$$

The asserted signs are the signs of $\lambda_i\mu_j$. □

Corollary 12.7 (Kronecker monotonicity). If

$$\mathbf{A} \succeq \mathbf{B} \succeq \mathbf{0}, \quad \mathbf{C} \succeq \mathbf{D} \succeq \mathbf{0},$$

then

$$\mathbf{A} \otimes \mathbf{C} \succeq \mathbf{B} \otimes \mathbf{D}.$$

Origin: H Ex. 16.8

Proof.

$$\mathbf{A} \otimes \mathbf{C} - \mathbf{B} \otimes \mathbf{D} = (\mathbf{A} - \mathbf{B}) \otimes \mathbf{C} + \mathbf{B} \otimes (\mathbf{C} - \mathbf{D}) \succeq \mathbf{0}.$$

□

1.3 Kronecker sums

Definition 12.8 (Kronecker sum). For $\mathbf{A} \in \mathbb{R}^{m \times m}$ and $\mathbf{B} \in \mathbb{R}^{n \times n}$, define

$$\mathbf{A} \boxplus \mathbf{B} := \mathbf{A} \otimes \mathbf{I}_n + \mathbf{I}_m \otimes \mathbf{B} \in \mathbb{R}^{mn \times mn}.$$

The symbol $\boxplus$ is reserved here for this matrix operation; it is not a direct sum of vector spaces.

Origin: Course formulation from S Ex. 4.3

Theorem 12.9 (Spectrum of a Kronecker sum). If $\lambda_1, \dots, \lambda_m$ and $\mu_1, \dots, \mu_n$ are the eigenvalues of $\mathbf{A}$ and $\mathbf{B}$, then the eigenvalues of $\mathbf{A} \boxplus \mathbf{B}$ are

$$\lambda_i + \mu_j, \quad 1 \leq i \leq m, \quad 1 \leq j \leq n.$$

If $\mathbf{A}$ and $\mathbf{B}$ are real symmetric with orthonormal eigenvectors $\mathbf{q}_i$ and $\mathbf{u}_j$, then $\mathbf{q}_i \otimes \mathbf{u}_j$ is an orthonormal eigenbasis of $\mathbf{A} \boxplus \mathbf{B}$.

Origin: Course formulation from S Ex. 4.3

Proof. With the Schur notation from Theorem 12.5,

$$(\mathbf{Q} \otimes \mathbf{R})^* (\mathbf{A} \boxplus \mathbf{B}) (\mathbf{Q} \otimes \mathbf{R}) = \mathbf{T} \otimes \mathbf{I}_n + \mathbf{I}_m \otimes \mathbf{S}.$$

The right side is upper triangular with diagonal $\lambda_i + \mu_j$. In the symmetric case,

$$\begin{aligned}(\mathbf{A} \boxplus \mathbf{B})(\mathbf{q}_i \otimes \mathbf{u}_j) &= \mathbf{A}\mathbf{q}_i \otimes \mathbf{u}_j + \mathbf{q}_i \otimes \mathbf{B}\mathbf{u}_j \\ &= (\lambda_i + \mu_j)(\mathbf{q}_i \otimes \mathbf{u}_j).\end{aligned}$$

Orthonormality was verified in Theorem 12.6. □

Corollary 12.10 (Definiteness of a Kronecker sum). If $\mathbf{A} \succeq \mathbf{O}$ and $\mathbf{B} \succeq \mathbf{O}$, then $\mathbf{A} \boxplus \mathbf{B} \succeq \mathbf{O}$. If, in addition, at least one factor is positive definite, then $\mathbf{A} \boxplus \mathbf{B} \succ \mathbf{O}$.

Origin: Course formulation from S Ex. 4.3

Proof. Every eigenvalue $\lambda_i + \mu_j$ is nonnegative. If all λ_i or all μ_j are positive, then every sum is positive. □

Example 12.2 (A rectangular-grid penalty). For $k \geq 2$, let $\mathbf{E}_k \in \mathbb{R}^{(k-1) \times k}$ have row i equal to $\mathbf{e}'_i - \mathbf{e}'_{i+1}$, and put $\mathbf{L}_k = \mathbf{E}'_k \mathbf{E}_k$. Then

$$\mathbf{x}' \mathbf{L}_k \mathbf{x} = \sum_{i=1}^{k-1} (x_i - x_{i+1})^2, \quad \text{null}(\mathbf{L}_k) = \text{span}\{\mathbf{1}_k\}.$$

Hence $\mathbf{L}_m \boxplus \mathbf{L}_n \succeq \mathbf{0}$ and

$$\text{null}(\mathbf{L}_m \boxplus \mathbf{L}_n) = \text{span}\{\mathbf{1}_m \otimes \mathbf{1}_n\}.$$

Indeed, a zero eigenvalue of the sum requires a zero eigenvalue from each factor.

Origin: Course formulation from S Ex. 4.3

2 The vec Operator and Matrix Equations

2.1 Column stacking and Roth's identity

Definition 12.11 (The vec operator). For $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_n) \in \mathbb{R}^{m \times n}$, define

$$\text{vec } \mathbf{X} := \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_n \end{pmatrix} \in \mathbb{R}^{mn}.$$

Thus x_{ij} occupies position $(j-1)m + i$. Column stacking is a linear bijection from $\mathbb{R}^{m \times n}$ to $\mathbb{R}^{mn}$.

Origin: A&M §10.2; H §16.2, Eq. (16.2.1); MC §2.4

Theorem 12.12 (Coordinate and inner-product identities). For matrices of the indicated orders,

$$\text{vec}(\alpha \mathbf{X} + \beta \mathbf{Y}) = \alpha \text{vec } \mathbf{X} + \beta \text{vec } \mathbf{Y}, \quad (12)$$

$$\text{vec}(\mathbf{a}\mathbf{b}') = \mathbf{b} \otimes \mathbf{a}, \quad (13)$$

$$(\text{vec } \mathbf{X})' \text{vec } \mathbf{Y} = \text{tr}(\mathbf{X}'\mathbf{Y}), \quad (14)$$

$$\|\text{vec } \mathbf{X}\|_2 = \|\mathbf{X}\|_F. \quad (15)$$

Origin: A&M Exs. 10.16, 10.18(a), 10.20(a); H Eqs. (16.2.3)–(16.2.6), (16.2.14); MC Ex. 3 and Prop. 5

Proof. Equation (12) holds columnwise. The j th column of $\mathbf{ab}'$ is $b_j\mathbf{a}$, so

$$\text{vec}(\mathbf{ab}') = \begin{pmatrix} b_1\mathbf{a} \\ \vdots \\ b_n\mathbf{a} \end{pmatrix} = \mathbf{b} \otimes \mathbf{a}.$$

Finally,

$$(\text{vec } \mathbf{X})' \text{vec } \mathbf{Y} = \sum_{i=1}^m \sum_{j=1}^n x_{ij}y_{ij} = \text{tr}(\mathbf{X}'\mathbf{Y}).$$

Set $\mathbf{Y} = \mathbf{X}$ and take the nonnegative square root to obtain (15). □

Theorem 12.13 (Roth's identity). If $\mathbf{A} \in \mathbb{R}^{r \times m}$, $\mathbf{X} \in \mathbb{R}^{m \times n}$, and $\mathbf{B} \in \mathbb{R}^{n \times s}$, then

$$\text{vec}(\mathbf{AXB}) = (\mathbf{B}' \otimes \mathbf{A}) \text{vec } \mathbf{X}. \quad (16)$$

In particular,

$$\text{vec}(\mathbf{AX}) = (\mathbf{I}_n \otimes \mathbf{A}) \text{vec } \mathbf{X}, \quad \text{vec}(\mathbf{XB}) = (\mathbf{B}' \otimes \mathbf{I}_m) \text{vec } \mathbf{X}.$$

Origin: A&M Ex. 10.18(b); H Thm. 16.2.1; MC Prop. 6

Proof. Write $\mathbf{X} = \sum_{j=1}^n \mathbf{x}_j \mathbf{e}_j'$. By (13) and the mixed-product rule,

$$\begin{aligned} \text{vec}(\mathbf{AXB}) &= \sum_{j=1}^n \text{vec}\{(\mathbf{Ax}_j)(\mathbf{B}'\mathbf{e}_j)'\} \\ &= \sum_{j=1}^n (\mathbf{B}'\mathbf{e}_j) \otimes (\mathbf{Ax}_j) \\ &= (\mathbf{B}' \otimes \mathbf{A}) \sum_{j=1}^n (\mathbf{e}_j \otimes \mathbf{x}_j) = (\mathbf{B}' \otimes \mathbf{A}) \text{vec } \mathbf{X}. \end{aligned}$$

The two special cases set $\mathbf{B} = \mathbf{I}_n$ or $\mathbf{A} = \mathbf{I}_m$. □

Corollary 12.14 (Trace of a four-factor product). For $\mathbf{A} \in \mathbb{R}^{r \times m}$, $\mathbf{X} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times s}$, and $\mathbf{Y} \in \mathbb{R}^{r \times s}$,

$$\text{tr}(\mathbf{Y}'\mathbf{A}\mathbf{X}\mathbf{B}) = (\text{vec } \mathbf{Y})'(\mathbf{B}' \otimes \mathbf{A}) \text{vec } \mathbf{X}.$$

Origin: A&M Ex. 10.20(b); H Thm. 16.2.2

Proof.

$$\text{tr}(\mathbf{Y}'\mathbf{A}\mathbf{X}\mathbf{B}) = (\text{vec } \mathbf{Y})' \text{vec}(\mathbf{A}\mathbf{X}\mathbf{B}) = (\text{vec } \mathbf{Y})'(\mathbf{B}' \otimes \mathbf{A}) \text{vec } \mathbf{X}.$$

□

2.2 Linear matrix operators and Sylvester equations

Definition 12.15 (A linear matrix operator). Fix $\mathbf{A}_l \in \mathbb{R}^{r \times m}$ and $\mathbf{B}_l \in \mathbb{R}^{n \times s}$ for $1 \leq l \leq k$. Define

$$\mathcal{L}(\mathbf{X}) := \sum_{l=1}^k \mathbf{A}_l \mathbf{X} \mathbf{B}_l, \quad \mathbf{M}_{\mathcal{L}} := \sum_{l=1}^k (\mathbf{B}_l' \otimes \mathbf{A}_l) \in \mathbb{R}^{rs \times mn}.$$

For real matrices of the same order, define the **Frobenius inner product** by

$$\langle \mathbf{X}, \mathbf{Y} \rangle_F := \text{tr}(\mathbf{X}'\mathbf{Y}).$$

A linear map $\mathcal{G} : \mathbb{R}^{r \times s} \rightarrow \mathbb{R}^{m \times n}$ is a **Frobenius adjoint** of $\mathcal{L}$ if

$$\langle \mathcal{L}(\mathbf{X}), \mathbf{Y} \rangle_F = \langle \mathbf{X}, \mathcal{G}(\mathbf{Y}) \rangle_F \quad (\mathbf{X} \in \mathbb{R}^{m \times n}, \mathbf{Y} \in \mathbb{R}^{r \times s}).$$

When exactly one such map exists, denote it by $\mathcal{L}^*$.

Origin: Course formulation from A&M Ex. 10.20(a); H Eqs. (16.2.14), (16.5.7)–(16.5.8)

Theorem 12.16 (Coordinate reduction of a matrix equation). For $\mathbf{X} \in \mathbb{R}^{m \times n}$ and $\mathbf{C} \in \mathbb{R}^{r \times s}$,

$$\text{vec } \mathcal{L}(\mathbf{X}) = \mathbf{M}_{\mathcal{L}} \text{vec } \mathbf{X}, \tag{17}$$

$$\mathcal{L}(\mathbf{X}) = \mathbf{C} \iff \mathbf{M}_{\mathcal{L}} \text{vec } \mathbf{X} = \text{vec } \mathbf{C}. \tag{18}$$

The equation is consistent exactly when $\text{vec } \mathbf{C} \in \text{col}(\mathbf{M}_{\mathcal{L}})$. Its unique minimum-Frobenius-norm

least-squares solution satisfies

$$\text{vec } \widehat{\mathbf{X}} = \mathbf{M}_{\mathcal{L}}^+ \text{vec } \mathbf{C}. \quad (19)$$

If $rs = mn$, every $\mathbf{C}$ has a unique solution exactly when $\mathbf{M}_{\mathcal{L}}$ is nonsingular.

Origin: A&M Exs. 10.45–10.49; H §16.5, Eqs. (16.5.7)–(16.5.9)

Proof. Roth's identity gives

$$\text{vec } \mathcal{L}(\mathbf{X}) = \sum_{l=1}^k (\mathbf{B}_l' \otimes \mathbf{A}_l) \text{vec } \mathbf{X} = \mathbf{M}_{\mathcal{L}} \text{vec } \mathbf{X}.$$

The consistency and uniqueness claims are the corresponding facts for a linear system. The Moore–Penrose formula minimizes $\|\mathbf{M}_{\mathcal{L}}\mathbf{x} - \text{vec } \mathbf{C}\|_2$ and then $\|\mathbf{x}\|_2$; the vec isometry converts these norms to Frobenius norms. $\square$

Theorem 12.17 (Sylvester equation). Let $\mathbf{A} \in \mathbb{R}^{m \times m}$ and $\mathbf{B} \in \mathbb{R}^{n \times n}$. The equation

$$\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B} = \mathbf{C} \quad (20)$$

has coefficient matrix

$$\mathbf{S}_{\mathbf{A},\mathbf{B}} := \mathbf{I}_n \otimes \mathbf{A} + \mathbf{B}' \otimes \mathbf{I}_m. \quad (21)$$

Its eigenvalues are

$$\lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B}), \quad 1 \leq i \leq m, \quad 1 \leq j \leq n.$$

Thus (20) has a unique solution for every $\mathbf{C}$ if and only if

$$\lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B}) \neq 0 \quad \text{for every } i, j. \quad (22)$$

Origin: Course formulation from H Eq. (16.5.8); S Ex. 3.23

Proof. Roth's identity gives

$$\text{vec}(\mathbf{A}\mathbf{X} + \mathbf{X}\mathbf{B}) = (\mathbf{I}_n \otimes \mathbf{A} + \mathbf{B}' \otimes \mathbf{I}_m) \text{vec } \mathbf{X}.$$

The coefficient is $\mathbf{B}' \boxplus \mathbf{A}$; transposition preserves the eigenvalues of $\mathbf{B}$. Theorem 12.9 gives its spectrum. A square linear system has a unique solution for every right side if and only if zero is not an eigenvalue of its coefficient matrix. $\square$

Corollary 12.18 (Symmetric Sylvester equation). Suppose

$$\mathbf{A} = \mathbf{Q} \operatorname{diag}(\lambda_1, \dots, \lambda_m) \mathbf{Q}', \quad \mathbf{B} = \mathbf{R} \operatorname{diag}(\mu_1, \dots, \mu_n) \mathbf{R}'$$

are symmetric and satisfy (22). Put $\mathbf{Y} = \mathbf{Q}' \mathbf{C} \mathbf{R}$ and

$$z_{ij} := \frac{y_{ij}}{\lambda_i + \mu_j}.$$

Then the unique solution is $\mathbf{X} = \mathbf{Q} \mathbf{Z} \mathbf{R}'$.

Origin: Course formulation from S Ex. 3.23

Proof. Premultiplication by $\mathbf{Q}'$ and postmultiplication by $\mathbf{R}$ transform (20) into

$$\operatorname{diag}(\lambda_i) \mathbf{Z} + \mathbf{Z} \operatorname{diag}(\mu_j) = \mathbf{Y}, \quad (\lambda_i + \mu_j) z_{ij} = y_{ij}.$$



2.3 Deterministic weighted matrix least squares

Definition 12.19 (Separable weighted residual norm). Let $\mathbf{A} \in \mathbb{R}^{n \times p}$ have full column rank, $\mathbf{Y} \in \mathbb{R}^{n \times q}$, $\mathbf{W} \succ \mathbf{O}$ of order n , and $\mathbf{R} \succ \mathbf{O}$ of order q . For $\mathbf{B} \in \mathbb{R}^{p \times q}$ define

$$\begin{aligned}\Phi(\mathbf{B}) &:= \text{tr}\{\mathbf{R}(\mathbf{Y} - \mathbf{AB})'\mathbf{W}(\mathbf{Y} - \mathbf{AB})\} \\ &= \|\mathbf{W}^{1/2}(\mathbf{Y} - \mathbf{AB})\mathbf{R}^{1/2}\|_F^2.\end{aligned}\tag{23}$$

Origin: Course formulation from CLM Def. 2.1

Theorem 12.20 (Two-sided weighting). The unique minimizer of (23) is

$$\widehat{\mathbf{B}} = (\mathbf{A}'\mathbf{WA})^{-1}\mathbf{A}'\mathbf{WY}.\tag{24}$$

It does not depend on the positive definite column weight $\mathbf{R}$.

Origin: Course formulation from H Thm. 16.2.1; CLM Thm. 2.2

Proof. Set

$$\tilde{\mathbf{A}} = \mathbf{W}^{1/2}\mathbf{A}, \quad \mathbf{Z} = \mathbf{W}^{1/2}\mathbf{Y}\mathbf{R}^{1/2}, \quad \mathbf{G} = \mathbf{B}\mathbf{R}^{1/2}.$$

Since $\mathbf{R}^{1/2}$ is nonsingular, $\mathbf{B} \mapsto \mathbf{G}$ is a bijection, and

$$\Phi(\mathbf{B}) = \|\mathbf{Z} - \tilde{\mathbf{A}}\mathbf{G}\|_F^2 = \sum_{j=1}^q \|\mathbf{z}_j - \tilde{\mathbf{A}}\mathbf{g}_j\|_2^2.$$

The full-column-rank least-squares formula applied to each column gives

$$\begin{aligned} \hat{\mathbf{G}} &= (\tilde{\mathbf{A}}'\tilde{\mathbf{A}})^{-1}\tilde{\mathbf{A}}'\mathbf{Z} \\ &= (\mathbf{A}'\mathbf{W}\mathbf{A})^{-1}\mathbf{A}'\mathbf{W}\mathbf{Y}\mathbf{R}^{1/2}. \end{aligned}$$

Right multiplication by $\mathbf{R}^{-1/2}$ gives (24). Positive definiteness of $\tilde{\mathbf{A}}'\tilde{\mathbf{A}}$ gives uniqueness. □

Corollary 12.21 (Vector form of two-sided weighting). With $\mathbf{b} = \text{vec } \mathbf{B}$ and $\mathbf{y} = \text{vec } \mathbf{Y}$,

$$\Phi(\mathbf{B}) = \|(\mathbf{R}^{1/2} \otimes \mathbf{W}^{1/2})\{\mathbf{y} - (\mathbf{I}_q \otimes \mathbf{A})\mathbf{b}\}\|_2^2, \quad (25)$$

$$\{\mathbf{R} \otimes (\mathbf{A}'\mathbf{W}\mathbf{A})\}\hat{\mathbf{b}} = (\mathbf{R} \otimes \mathbf{A}'\mathbf{W})\mathbf{y}. \quad (26)$$

Origin: Course formulation from H Eqs. (16.1.19), (16.2.10); CLM Thm. 2.1

Proof. Roth's identity and the vec isometry give (25). Its design matrix is

$$(\mathbf{R}^{1/2} \otimes \mathbf{W}^{1/2})(\mathbf{I}_q \otimes \mathbf{A}) = \mathbf{R}^{1/2} \otimes (\mathbf{W}^{1/2}\mathbf{A}).$$

The ordinary normal equations and the mixed-product rule give (26). □

3 Commutation and Symmetric Coordinates

3.1 The commutation matrix

Definition 12.22 (Commutation matrix). For positive integers m, n , the **commutation matrix** $\mathbf{K}_{mn} \in \mathbb{R}^{mn \times mn}$ is the unique permutation matrix such that

$$\mathbf{K}_{mn} \text{vec } \mathbf{X} = \text{vec } \mathbf{X}' \quad \text{for every } \mathbf{X} \in \mathbb{R}^{m \times n}. \quad (27)$$

The order of the indices records the order of $\mathbf{X}$.

Origin: A&M §11.1; H §16.3a, Eq. (16.3.1); MC §2.5

Theorem 12.23 (Commutation identities). The commutation matrix satisfies

$$\mathbf{K}'_{mn} = \mathbf{K}_{mn}^{-1} = \mathbf{K}_{nm}, \quad (28)$$

$$\text{tr } \mathbf{K}_{nn} = n. \quad (29)$$

If $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{p \times q}$, then

$$\mathbf{K}_{pm}(\mathbf{A} \otimes \mathbf{B}) = (\mathbf{B} \otimes \mathbf{A})\mathbf{K}_{qn}, \quad (30)$$

$$\mathbf{B} \otimes \mathbf{A} = \mathbf{K}_{pm}(\mathbf{A} \otimes \mathbf{B})\mathbf{K}_{nq}. \quad (31)$$

Origin: A&M Exs. 11.1, 11.3–11.4, 11.13; H Eqs. (16.3.6), (16.3.8) and Thm. 16.3.2; MC Props. 7–8

Proof. For every $\mathbf{X} \in \mathbb{R}^{m \times n}$,

$$\mathbf{K}_{nm} \mathbf{K}_{mn} \text{vec } \mathbf{X} = \mathbf{K}_{nm} \text{vec } \mathbf{X}' = \text{vec } \mathbf{X}.$$

Thus $\mathbf{K}_{nm} = \mathbf{K}_{mn}^{-1}$. A permutation matrix is orthogonal, so $\mathbf{K}'_{mn} = \mathbf{K}_{mn}^{-1}$. Exactly the n diagonal coordinates of an $n \times n$ matrix retain their positions under transposition, giving $\text{tr } \mathbf{K}_{nn} = n$.

For arbitrary $\mathbf{X} \in \mathbb{R}^{q \times n}$, Roth's identity gives

$$\begin{aligned} \mathbf{K}_{pm}(\mathbf{A} \otimes \mathbf{B}) \text{vec } \mathbf{X} &= \text{vec}\{(\mathbf{B}\mathbf{X}\mathbf{A}')'\} \\ &= \text{vec}(\mathbf{A}\mathbf{X}'\mathbf{B}') \\ &= (\mathbf{B} \otimes \mathbf{A}) \text{vec } \mathbf{X}' = (\mathbf{B} \otimes \mathbf{A}) \mathbf{K}_{qn} \text{vec } \mathbf{X}. \end{aligned}$$

The bijectivity of vec gives (30); right multiplication by $\mathbf{K}_{nq} = \mathbf{K}_{qn}^{-1}$ gives (31). □

Example 12.3 (The matrix $\mathbf{K}_{23}$).

$$\mathbf{K}_{23} = \begin{pmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{K}_{23} \begin{pmatrix} x_{11} \\ x_{21} \\ x_{12} \\ x_{22} \\ x_{13} \\ x_{23} \end{pmatrix} = \begin{pmatrix} x_{11} \\ x_{12} \\ x_{13} \\ x_{21} \\ x_{22} \\ x_{23} \end{pmatrix}.$$

Thus $\mathbf{K}'_{23} = \mathbf{K}_{32}$; $\mathbf{K}_{23}$ need not be symmetric.

Origin: A&M Ex. 11.9; H §16.3a

3.2 Symmetric and skew-symmetric coordinates

Definition 12.24 (Symmetrizer and skew-symmetrizer). For $n \geq 1$, define

$$\mathbf{N}_n := \frac{1}{2}(\mathbf{I}_{n^2} + \mathbf{K}_{nn}), \quad \mathbf{M}_n := \frac{1}{2}(\mathbf{I}_{n^2} - \mathbf{K}_{nn}).$$

Origin: A&M §§11.1–11.2; H §16.4c, Eq. (16.4.19); MC Ex. 5

Theorem 12.25 (Two orthogonal coordinate projections). For $\mathbf{X} \in \mathbb{R}^{n \times n}$,

$$\mathbf{N}_n \operatorname{vec} \mathbf{X} = \operatorname{vec} \frac{\mathbf{X} + \mathbf{X}'}{2}, \quad (32)$$

$$\mathbf{M}_n \operatorname{vec} \mathbf{X} = \operatorname{vec} \frac{\mathbf{X} - \mathbf{X}'}{2}. \quad (33)$$

Moreover,

$$\mathbf{N}'_n = \mathbf{N}_n^2 = \mathbf{N}_n, \quad \mathbf{M}'_n = \mathbf{M}_n^2 = \mathbf{M}_n, \quad \mathbf{N}_n \mathbf{M}_n = \mathbf{O}, \quad \mathbf{N}_n + \mathbf{M}_n = \mathbf{I}_{n^2}, \quad (34)$$

$$\operatorname{rank} \mathbf{N}_n = \frac{n(n+1)}{2}, \quad \operatorname{rank} \mathbf{M}_n = \frac{n(n-1)}{2}. \quad (35)$$

Thus $\text{col}(\mathbf{N}_n)$ and $\text{col}(\mathbf{M}_n)$ are respectively the vec images of the symmetric and skew-symmetric $n \times n$ matrices.

Origin: A&M Exs. 11.13–11.14, 11.16–11.17; H Eqs. (16.4.19)–(16.4.21); MC Ex. 5

Proof. Equations (32)–(33) use $\mathbf{K}_{nn} \text{vec } \mathbf{X} = \text{vec } \mathbf{X}'$. Since $\mathbf{K}'_{nn} = \mathbf{K}_{nn}$ and $\mathbf{K}_{nn}^2 = \mathbf{I}_{n^2}$,

$$\begin{aligned}\mathbf{N}_n^2 &= \frac{1}{4}(\mathbf{I} + 2\mathbf{K}_{nn} + \mathbf{K}_{nn}^2) = \mathbf{N}_n, \\ \mathbf{M}_n^2 &= \frac{1}{4}(\mathbf{I} - 2\mathbf{K}_{nn} + \mathbf{K}_{nn}^2) = \mathbf{M}_n, \\ \mathbf{N}_n \mathbf{M}_n &= \frac{1}{4}(\mathbf{I} - \mathbf{K}_{nn}^2) = \mathbf{O}.\end{aligned}$$

For a symmetric idempotent, rank equals trace. Hence

$$\begin{aligned}\text{rank } \mathbf{N}_n &= \text{tr } \mathbf{N}_n = \frac{1}{2}(n^2 + n), \\ \text{rank } \mathbf{M}_n &= \text{tr } \mathbf{M}_n = \frac{1}{2}(n^2 - n).\end{aligned}$$

The action formulas identify the two ranges. □

Corollary 12.26 (Commutation spectrum). The eigenvalues of $\mathbf{K}_{nn}$ are 1 and -1 , with multiplicities $n(n+1)/2$ and $n(n-1)/2$. Therefore

$$\det \mathbf{K}_{nn} = (-1)^{n(n-1)/2}.$$

Origin: A&M Ex. 11.13

Proof. From $\mathbf{K}_{nn} = \mathbf{N}_n - \mathbf{M}_n$, its restrictions to $\text{col}(\mathbf{N}_n)$ and $\text{col}(\mathbf{M}_n)$ are $\mathbf{I}$ and $-\mathbf{I}$. Their dimensions are given by (35). $\square$

3.3 vech, elimination, duplication, and congruence

Definition 12.27 (vech, elimination, and duplication). For $\mathbf{X} = (x_{ij}) \in \mathbb{R}^{n \times n}$, $\text{vech}(\mathbf{X})$ stacks the entries x_{ij} with $i \geq j$, column by column:

$$\text{vech}(\mathbf{X}) = (x_{11}, x_{21}, \dots, x_{n1}, x_{22}, \dots, x_{n2}, \dots, x_{nn})'.$$

Put $d_n = n(n+1)/2$. The **elimination matrix** $\mathbf{L}_n \in \mathbb{R}^{d_n \times n^2}$ and the **duplication matrix**

$\mathbf{D}_n \in \mathbb{R}^{n^2 \times d_n}$ are defined by

$$\mathbf{L}_n \text{vec } \mathbf{X} = \text{vech}(\mathbf{X}) \quad (\mathbf{X} \in \mathbb{R}^{n \times n}), \quad (36)$$

$$\mathbf{D}_n \text{vech}(\mathbf{S}) = \text{vec } \mathbf{S} \quad (\mathbf{S} = \mathbf{S}'). \quad (37)$$

Origin: A&M §11.3; H §§16.4a–16.4b and Ex. 16.25; MC §2.6

Example 12.4 (Elimination is not averaging). For $n = 2$,

$$\mathbf{L}_2 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{D}_2 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix},$$

whereas

$$\mathbf{D}_2^+ = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}.$$

Thus $\mathbf{L}_2 \text{vec } \mathbf{X} = (x_{11}, x_{21}, x_{22})'$, while $\mathbf{D}_2^+ \text{vec } \mathbf{X} = (x_{11}, (x_{12} + x_{21})/2, x_{22})'$.

Origin: A&M Exs. 11.27, 11.29; H Eqs. (16.4.8a) and (16.4.13)

Theorem 12.28 (Duplication and the symmetric projector). The matrix $\mathbf{D}_n$ has full column rank and

$$\mathbf{L}_n \mathbf{D}_n = \mathbf{I}_{d_n}, \quad (38)$$

$$\mathbf{D}_n^+ = (\mathbf{D}_n' \mathbf{D}_n)^{-1} \mathbf{D}_n', \quad (39)$$

$$\mathbf{D}_n^+ \mathbf{D}_n = \mathbf{I}_{d_n}, \quad (40)$$

$$\mathbf{D}_n \mathbf{D}_n^+ = \mathbf{N}_n, \quad (41)$$

$$\mathbf{D}_n^+ = \mathbf{L}_n \mathbf{N}_n. \quad (42)$$

The columns of $\mathbf{D}_n$ are mutually orthogonal. In $\mathbf{D}_n' \mathbf{D}_n$, the coordinate for s_{ii} has diagonal weight 1 and the coordinate for s_{ij} , $i > j$, has diagonal weight 2.

Origin: A&M Exs. 11.26–11.30; H Eqs. (16.4.4), (16.4.13), (16.4.22); MC Prop. 9

Proof. Every column of $\mathbf{D}_n$ contains either one 1 for a diagonal entry or two 1s for a mirrored off-diagonal pair; distinct columns have disjoint supports. Hence $\mathbf{D}_n$ has full column rank and the stated form of $\mathbf{D}'_n \mathbf{D}_n$ holds. Equations (39) and (40) are the full-column-rank Moore–Penrose formula. Equation (38) holds because $\mathbf{L}_n$ deletes the duplicated upper-triangular coordinate. The column space of $\mathbf{D}_n$ is the vec image of the symmetric matrices, which is $\text{col}(\mathbf{N}_n)$. Both $\mathbf{D}_n \mathbf{D}_n^+$ and $\mathbf{N}_n$ are orthogonal projectors onto this space, proving (41). For arbitrary $\mathbf{X}$,

$$\mathbf{D}_n^+ \text{vec } \mathbf{X} = \text{vech } \frac{\mathbf{X} + \mathbf{X}'}{2} = \mathbf{L}_n \mathbf{N}_n \text{vec } \mathbf{X},$$

which proves (42) by the bijectivity of vec. □

Definition 12.29 (Congruence coordinate matrix). For $\mathbf{A} \in \mathbb{R}^{r \times n}$ define

$$\mathbf{C}_{\mathbf{A}} := \mathbf{D}_r^+(\mathbf{A} \otimes \mathbf{A})\mathbf{D}_n \in \mathbb{R}^{d_r \times d_n}, \quad d_k = \frac{k(k+1)}{2}.$$

Origin: H Eq. (16.4.25); A&M Ex. 11.31; MC Prop. 10

Theorem 12.30 (Congruence in symmetric coordinates). For every symmetric $\mathbf{X} \in \mathbb{R}^{n \times n}$,

$$\text{vech}(\mathbf{A}\mathbf{X}\mathbf{A}') = \mathbf{C}_{\mathbf{A}} \text{vech}(\mathbf{X}). \quad (43)$$

If $\mathbf{A} \in \mathbb{R}^{r \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$, then

$$\mathbf{C}_{\mathbf{A}}\mathbf{C}_{\mathbf{B}} = \mathbf{C}_{\mathbf{AB}}. \quad (44)$$

For square $\mathbf{A}$ of order n and $t = \text{rank } \mathbf{A}$,

$$\text{rank } \mathbf{C}_{\mathbf{A}} = \frac{t(t+1)}{2}, \quad (45)$$

$$\mathbf{C}_{\mathbf{A}}^{-1} = \mathbf{C}_{\mathbf{A}^{-1}} \quad (\mathbf{A} \text{ nonsingular}), \quad (46)$$

$$\det \mathbf{C}_{\mathbf{A}} = (\det \mathbf{A})^{n+1}. \quad (47)$$

Origin: A&M Exs. 11.31–11.33; H Thms. 16.4.1–16.4.2

Proof. Roth's identity and the symmetry of $\mathbf{A}\mathbf{X}\mathbf{A}'$ give

$$\mathbf{C}_{\mathbf{A}} \text{vech } \mathbf{X} = \mathbf{D}_r^+(\mathbf{A} \otimes \mathbf{A}) \text{vec } \mathbf{X} = \mathbf{D}_r^+ \text{vec}(\mathbf{A}\mathbf{X}\mathbf{A}') = \text{vech}(\mathbf{A}\mathbf{X}\mathbf{A}').$$

Applying this identity twice to arbitrary symmetric $\mathbf{X}$ proves (44); the map $\mathbf{X} \mapsto \text{vech } \mathbf{X}$ is bijective on symmetric matrices. Taking $\mathbf{B} = \mathbf{A}^{-1}$ proves (46).

Let $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}'$ be an SVD with t positive singular values. The matrices $\mathbf{C}_{\mathbf{U}}$ and $\mathbf{C}_{\mathbf{V}'}$ are nonsingular by (46), while $\mathbf{X} \mapsto \mathbf{\Sigma}\mathbf{X}\mathbf{\Sigma}'$ has as its range the symmetric matrices supported on a $t \times t$ principal block. Its range has dimension $t(t+1)/2$, proving (45).

If $\mathbf{A}$ is singular, both sides of (47) are zero. For the determinant calculation, extend $\mathbf{C}_{\mathbf{A}} = \mathbf{D}_n^+(\mathbf{A} \otimes \mathbf{A})\mathbf{D}_n$ complex-linearly; the preceding composition identity is algebraic and is unchanged. If $\mathbf{A}$ is nonsingular, complex Schur triangularisation gives $\mathbf{A} = \mathbf{STS}^{-1}$. By (44),

$$\mathbf{C}_{\mathbf{A}} = \mathbf{C}_{\mathbf{S}}\mathbf{C}_{\mathbf{T}}\mathbf{C}_{\mathbf{S}}^{-1}.$$

For $i \geq j$, let

$$\mathbf{F}_{ij} := \begin{cases} \mathbf{e}_i \mathbf{e}'_i, & i = j, \\ \mathbf{e}_i \mathbf{e}'_j + \mathbf{e}_j \mathbf{e}'_i, & i > j. \end{cases}$$

In the basis ordered as vech, the map $\mathbf{X} \mapsto \mathbf{T}\mathbf{X}\mathbf{T}'$ is upper triangular: the coefficient of $\mathbf{F}_{ij}$ in $\mathbf{T}\mathbf{F}_{ij}\mathbf{T}'$ is $t_{ii}t_{jj}$. Hence

$$\det \mathbf{C}_{\mathbf{A}} = \det \mathbf{C}_{\mathbf{T}} = \prod_{i \geq j} t_{ii}t_{jj} = \prod_{i=1}^n t_{ii}^{n+1} = (\det \mathbf{A})^{n+1}.$$

□

Remark. The matrices $\mathbf{B}' \otimes \mathbf{A}$ and $\mathbf{C}_\mathbf{A}$ are coordinate matrices of the linear maps $\mathbf{X} \mapsto \mathbf{A}\mathbf{X}\mathbf{B}$ and $\mathbf{X} \mapsto \mathbf{A}\mathbf{X}\mathbf{A}'$. Thus unrestricted coordinates use $\mathbf{B}' \otimes \mathbf{A}$, whereas symmetric congruence coordinates use $\mathbf{C}_\mathbf{A}$.

4 Computation: Numerical Checks in R

4.1 Mixed products and inverses (§1.1)

```

1 A <- matrix(c(1, 3, 2, 4), 2, 2)
2 B <- matrix(c(2, -1, 0, 1), 2, 2)
3 C <- matrix(c(1, 0, 2, 1), 2, 2)
4 D <- matrix(c(3, 1, -1, 2), 2, 2)
5 mixed_left <- (A %x% B) %*% (C %x% D)
6 mixed_right <- (A %*% C) %x% (B %*% D)
7 inverse_manual <- solve(A) %x% solve(B)
8 inverse_package <- solve(A %x% B)
9 c(mixed_residual = max(abs(mixed_left - mixed_right)),
10   inverse_residual = max(abs(inverse_manual - inverse_package)))

```

```

##      mixed_residual inverse_residual
##              0              0

```

4.2 Spectrum, determinant, and rank (§1.2)

```

1 A <- matrix(c(4, 1, 1, 2), 2, 2)
2 B <- matrix(c(3, 0, 1, 0, 2, 0, 1, 0, 1), 3, 3)
3 ev_A <- eigen(A, symmetric = TRUE)$values
4 ev_B <- eigen(B, symmetric = TRUE)$values
5 ev_product <- sort(as.vector(outer(ev_A, ev_B, "*")))
6 ev_package <- sort(eigen(A %x% B, symmetric = TRUE)$values)
7 rank_tol <- 1e-10
8 rank_qr <- function(M, tol = rank_tol) qr(M, tol = tol)$rank
9 c(eigenvalue_residual = max(abs(ev_product - ev_package)),
10   determinant_residual = det(A %x% B) - det(A)^3 * det(B)^2,
11   rank_at_1e_10 = rank_qr(A %x% B),
12   rank_product_at_1e_10 = rank_qr(A) * rank_qr(B))

```

```

## eigenvalue_residual determinant_residual
## 8.882e-15 4.547e-12
## rank_at_1e_10 rank_product_at_1e_10
## 6.000e+00 6.000e+00

```

4.3 A Sylvester solve (§2.2)

```

1 A <- matrix(c(3, 1, 1, 2), 2, 2)
2 B <- matrix(c(2, -1, -1, 4), 2, 2)
3 C <- matrix(c(1, 2, -1, 3), 2, 2)
4 coef <- diag(2) %x% A + t(B) %x% diag(2)
5 X_vec <- matrix(solve(coef, c(C)), 2, 2)
6 ea <- eigen(A, symmetric = TRUE)
7 eb <- eigen(B, symmetric = TRUE)
8 Y <- t(ea$vectors) %*% C %*% eb$vectors
9 Z <- Y / outer(ea$values, eb$values, "+")
10 X_spectral <- ea$vectors %*% Z %*% t(eb$vectors)
11 c(equation_residual = max(abs(A %*% X_vec + X_vec %*% B - C)),
12   route_difference = max(abs(X_vec - X_spectral)))

```

```

## equation_residual  route_difference
##                4.441e-16          2.220e-16

```

4.4 Two-sided weighted least squares (§2.3)

```

1 A <- cbind(1, c(-2, -1, 1, 2))
2 Y <- matrix(c(1, 2, 0, 3, 2, -1, 1, 4), 4, 2)

```

```

3 W <- diag(c(1, 2, 3, 4))
4 R <- matrix(c(2, 1, 1, 3), 2, 2)
5 B_formula <- solve(t(A) %*% W %*% A, t(A) %*% W %*% Y)
6 normal_matrix <- R %x% (t(A) %*% W %*% A)
7 normal_rhs <- (R %x% (t(A) %*% W)) %*% c(Y)
8 B_vec <- matrix(solve(normal_matrix, normal_rhs), ncol(A), ncol(Y))
9 normal_residual <- t(A) %*% W %*% (Y - A %*% B_formula) %*% R
10 c(route_difference = max(abs(B_formula - B_vec)),
11    normal_equation_residual = max(abs(normal_residual)))

```

```

##           route_difference normal_equation_residual
##           2.220e-16           6.217e-15

```

4.5 Pattern matrices (§3)

```

1 commutation <- function(m, n) {
2   K <- matrix(0, m * n, m * n)
3   for (i in seq_len(m)) for (j in seq_len(n))
4     K[(i - 1) * n + j, (j - 1) * m + i] <- 1
5   K
6 }
7 elimination <- function(n) {

```

```
8   keep <- as.vector(row(matrix(0, n, n)) >= col(matrix(0, n, n)))
9   diag(n * n)[keep, , drop = FALSE]
10 }
11 duplication <- function(n) {
12   d <- n * (n + 1) / 2
13   D <- matrix(0, n * n, d); h <- 0
14   for (j in seq_len(n)) for (i in j:n) {
15     h <- h + 1
16     D[(j - 1) * n + i, h] <- 1
17     D[(i - 1) * n + j, h] <- 1
18   }
19   D
20 }
21 n <- 3; K <- commutation(n, n); I <- diag(n * n)
22 N <- (I + K) / 2; M <- (I - K) / 2
23 L <- elimination(n); D <- duplication(n)
24 Dplus <- solve(t(D) %*% D, t(D))
25 X <- matrix(1:9, n, n)
26 c(transpose_residual = max(abs(K %*% c(X) - c(t(X)))),
27   projector_residual = max(abs(N %*% N - N)),
28   orthogonality_residual = max(abs(N %*% M)),
```

```

29 duplication_residual = max(abs(D %**% Dplus - N)),
30 averaging_residual = max(abs(Dplus - L %**% N))

```

```

##      transpose_residual      projector_residual
##              0              0
## orthogonality_residual      duplication_residual
##              0              0
##      averaging_residual
##              0

```

4.6 Congruence in vech coordinates (§3.3)

```

1 n <- 3; D <- duplication(n); Dplus <- solve(t(D) %**% D, t(D))
2 A <- matrix(c(2, 0, 1, 1, 1, 0, 0, 1, 1), n, n)
3 X <- matrix(c(3, 1, 0, 1, 2, -1, 0, -1, 4), n, n)
4 C_A <- Dplus %**% (A %x% A) %**% D
5 vech <- function(S) elimination(n) %**% c(S)
6 c(coordinate_residual = max(abs(C_A %**% vech(X) - vech(A %**% X %**% t(A)))),
7   determinant_residual = det(C_A) - det(A)^(n + 1))

```

```

## coordinate_residual determinant_residual
##              0              0

```

5 Pitfalls

5.1 Reversing Kronecker factors

$$\mathbf{A} \otimes \mathbf{B} \neq \mathbf{B} \otimes \mathbf{A}, \quad \mathbf{B} \otimes \mathbf{A} = \mathbf{K}_{pm}(\mathbf{A} \otimes \mathbf{B})\mathbf{K}_{nq}.$$

The equality on the right requires $\mathbf{A}$ of order $m \times n$ and $\mathbf{B}$ of order $p \times q$; the two permutation matrices have different index orders.

5.2 Applying the mixed-product rule without conformability

$$(\mathbf{A}_{m \times n} \otimes \mathbf{B}_{p \times q})(\mathbf{C}_{u \times v} \otimes \mathbf{D}_{r \times s})$$

is defined in the mixed-product form only when $n = u$ and $q = r$. These are exactly the conditions that define $\mathbf{AC}$ and $\mathbf{BD}$.

5.3 Stacking rows instead of columns

$$\text{vec} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a, c, b, d)', \quad \text{vec} \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix}' \right\} = (a, b, c, d)'.$$

Row stacking silently inserts a commutation matrix into every Roth identity.

5.4 Changing the sign in a Sylvester coefficient

$$\begin{aligned}\mathbf{AX} + \mathbf{XB} = \mathbf{C} &\longleftrightarrow \{\mathbf{I} \otimes \mathbf{A} + \mathbf{B}' \otimes \mathbf{I}\} \operatorname{vec} \mathbf{X} = \operatorname{vec} \mathbf{C}, \\ \mathbf{AX} - \mathbf{XB} = \mathbf{C} &\longleftrightarrow \{\mathbf{I} \otimes \mathbf{A} - \mathbf{B}' \otimes \mathbf{I}\} \operatorname{vec} \mathbf{X} = \operatorname{vec} \mathbf{C}.\end{aligned}$$

The corresponding exclusion conditions are $\lambda_i(\mathbf{A}) + \lambda_j(\mathbf{B}) \neq 0$ and $\lambda_i(\mathbf{A}) - \lambda_j(\mathbf{B}) \neq 0$.

5.5 Replacing averaging by elimination

$$\mathbf{L}_n \operatorname{vec} \mathbf{X} = \operatorname{vech} \mathbf{X}, \quad \mathbf{D}_n^+ \operatorname{vec} \mathbf{X} = \operatorname{vech} \frac{\mathbf{X} + \mathbf{X}'}{2}.$$

Thus $\mathbf{L}_n = \mathbf{D}_n^+$ on vecs of symmetric matrices, but not on all of $\mathbb{R}^{n^2}$.

5.6 Expanding a structured system prematurely

$$\mathbf{AX} + \mathbf{XB} = \mathbf{C} \rightsquigarrow (\mathbf{I}_n \otimes \mathbf{A} + \mathbf{B}' \otimes \mathbf{I}_m) \operatorname{vec} \mathbf{X} = \operatorname{vec} \mathbf{C},$$

whose coefficient has $m^2 n^2$ entries. The identity specifies the operator; it does not require storing that coefficient when a matrix-equation solver is available.

6 Exercises

Exercise 12.1 (Kronecker products of vectors). For conformable vectors $\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$, prove

$$(\mathbf{a} \otimes \mathbf{b})(\mathbf{c} \otimes \mathbf{d})' = (\mathbf{a}\mathbf{c}') \otimes (\mathbf{b}\mathbf{d}'), \quad (\mathbf{a} \otimes \mathbf{b})'(\mathbf{c} \otimes \mathbf{d}) = (\mathbf{a}'\mathbf{c})(\mathbf{b}'\mathbf{d}).$$

Deduce $\|\mathbf{a} \otimes \mathbf{b}\|_2 = \|\mathbf{a}\|_2 \|\mathbf{b}\|_2$.

Origin: A&M Ex. 10.6; H Eqs. (16.1.2)–(16.1.4) and Ex. 16.9

Solution. Since $(\mathbf{c} \otimes \mathbf{d})' = \mathbf{c}' \otimes \mathbf{d}'$, the mixed-product rule gives

$$(\mathbf{a} \otimes \mathbf{b})(\mathbf{c}' \otimes \mathbf{d}') = (\mathbf{a}\mathbf{c}') \otimes (\mathbf{b}\mathbf{d}').$$

Apply the same rule to the scalar product. With $\mathbf{c} = \mathbf{a}$ and $\mathbf{d} = \mathbf{b}$,

$$\|\mathbf{a} \otimes \mathbf{b}\|_2^2 = (\mathbf{a}'\mathbf{a})(\mathbf{b}'\mathbf{b}) = \|\mathbf{a}\|_2^2 \|\mathbf{b}\|_2^2.$$



Exercise 12.2 (Identifying Kronecker factors). Suppose $\mathbf{A} \otimes \mathbf{B} = \mathbf{C} \otimes \mathbf{D} \neq \mathbf{O}$, where $\mathbf{A}, \mathbf{C}$ have the same order and $\mathbf{B}, \mathbf{D}$ have the same order. Prove that some $\alpha \neq 0$ satisfies

$$\mathbf{C} = \alpha \mathbf{A}, \quad \mathbf{D} = \alpha^{-1} \mathbf{B}.$$

Origin: Course formulation from S Cor. 4.1

Solution. Choose $a_{ij} \mathbf{B} \neq \mathbf{O}$; then $a_{ij} \neq 0$ and $\mathbf{B} \neq \mathbf{O}$. Equality of the (i, j) blocks gives

$$a_{ij} \mathbf{B} = c_{ij} \mathbf{D} \neq \mathbf{O},$$

so $c_{ij} \neq 0$ and $\mathbf{D} = (a_{ij}/c_{ij}) \mathbf{B}$. Substitution into every block gives

$$a_{kl} \mathbf{B} = c_{kl} (a_{ij}/c_{ij}) \mathbf{B}.$$

Because $\mathbf{B} \neq \mathbf{O}$, $c_{kl} = (c_{ij}/a_{ij}) a_{kl}$. Set $\alpha = c_{ij}/a_{ij}$. 

Exercise 12.3 (Kronecker products of projectors). Let $\mathbf{P} \in \mathbb{R}^{m \times m}$ and $\mathbf{Q} \in \mathbb{R}^{n \times n}$ satisfy $\mathbf{P} = \mathbf{P}' = \mathbf{P}^2$ and $\mathbf{Q} = \mathbf{Q}' = \mathbf{Q}^2$. Prove that $\mathbf{P} \otimes \mathbf{Q}$ is the orthogonal projector onto

$\text{span}\{\mathbf{p}_i \otimes \mathbf{q}_j : \mathbf{p}_i \text{ runs through a basis of } \text{col } \mathbf{P}, \mathbf{q}_j \text{ runs through a basis of } \text{col } \mathbf{Q}\}.$

Origin: A&M Ex. 10.7(c); H Exs. 16.4, 16.6

Solution. The transpose and mixed-product rules give

$$(\mathbf{P} \otimes \mathbf{Q})' = \mathbf{P} \otimes \mathbf{Q}, \quad (\mathbf{P} \otimes \mathbf{Q})^2 = \mathbf{P} \otimes \mathbf{Q}.$$

If $\mathbf{x} = \sum_i a_i \mathbf{p}_i$ and $\mathbf{y} = \sum_j b_j \mathbf{q}_j$, then $\mathbf{x} \otimes \mathbf{y}$ lies in the displayed span. For every displayed basis product,

$$(\mathbf{P} \otimes \mathbf{Q})(\mathbf{p}_i \otimes \mathbf{q}_j) = \mathbf{p}_i \otimes \mathbf{q}_j,$$

so the displayed span is contained in $\text{col}(\mathbf{P} \otimes \mathbf{Q})$. Conversely, every column of $\mathbf{P} \otimes \mathbf{Q}$ is a Kronecker product of a column of $\mathbf{P}$ and a column of $\mathbf{Q}$, hence lies in the displayed span. ■

Exercise 12.4 (A Kronecker order equivalence). For symmetric positive semidefinite $\mathbf{A}, \mathbf{B}$ of the same order, prove

$$\mathbf{A} \otimes \mathbf{A} \succeq \mathbf{B} \otimes \mathbf{B} \iff \mathbf{A} \succeq \mathbf{B}.$$

Give a 1×1 counterexample when positive semidefiniteness is omitted.

Origin: A&M Ex. 10.14

Solution. If $\mathbf{A} \succeq \mathbf{B}$, Corollary 12.7 applies. Conversely, for every $\mathbf{x}$,

$$\begin{aligned} 0 &\leq (\mathbf{x} \otimes \mathbf{x})'(\mathbf{A} \otimes \mathbf{A} - \mathbf{B} \otimes \mathbf{B})(\mathbf{x} \otimes \mathbf{x}) \\ &= (\mathbf{x}'\mathbf{A}\mathbf{x})^2 - (\mathbf{x}'\mathbf{B}\mathbf{x})^2 \\ &= \{\mathbf{x}'(\mathbf{A} + \mathbf{B})\mathbf{x}\}\{\mathbf{x}'(\mathbf{A} - \mathbf{B})\mathbf{x}\}. \end{aligned}$$

If the first factor is positive, the second is nonnegative. If it is zero, both $\mathbf{x}'\mathbf{A}\mathbf{x}$ and $\mathbf{x}'\mathbf{B}\mathbf{x}$ are zero, so the second factor is zero. Thus $\mathbf{x}'(\mathbf{A} - \mathbf{B})\mathbf{x} \geq 0$ for every $\mathbf{x}$. For $\mathbf{A} = (-2)$ and $\mathbf{B} = (1)$, $\mathbf{A} \otimes \mathbf{A} = (4) \succeq (1) = \mathbf{B} \otimes \mathbf{B}$, but $\mathbf{A} \not\succeq \mathbf{B}$. ■

Exercise 12.5 (Grid energy). Let $m, n \geq 2$ and use the matrices $\mathbf{E}_k, \mathbf{L}_k$ from Example 12.2. Index z_{ij} by $1 \leq i \leq m, 1 \leq j \leq n$, and stack the j index within the i index into $\mathbf{z} \in \mathbb{R}^{mn}$. Prove

$$\mathbf{z}'(\mathbf{L}_m \boxplus \mathbf{L}_n)\mathbf{z} = \sum_{i=1}^{m-1} \sum_{j=1}^n (z_{ij} - z_{i+1,j})^2 + \sum_{i=1}^m \sum_{j=1}^{n-1} (z_{ij} - z_{i,j+1})^2.$$

Characterize equality.

Origin: Course formulation from S Ex. 4.3

Solution. The two terms are

$$\begin{aligned} \mathbf{z}'(\mathbf{L}_m \otimes \mathbf{I}_n)\mathbf{z} &= \sum_{i=1}^{m-1} \sum_{j=1}^n (z_{ij} - z_{i+1,j})^2, \\ \mathbf{z}'(\mathbf{I}_m \otimes \mathbf{L}_n)\mathbf{z} &= \sum_{i=1}^m \sum_{j=1}^{n-1} (z_{ij} - z_{i,j+1})^2. \end{aligned}$$

Equality to zero holds exactly when all horizontal and vertical differences vanish, equivalently

$z_{ij} = c$ for all i, j , or $\mathbf{z} = c(\mathbf{1}_m \otimes \mathbf{1}_n)$. 

Exercise 12.6 (A commutator equation). Let $\mathbf{A} = \text{diag}(\lambda_1, \dots, \lambda_n)$ with distinct diagonal entries and $\mathbf{C} \in \mathbb{R}^{n \times n}$. Characterize all solutions of

$$\mathbf{A}\mathbf{X} - \mathbf{X}\mathbf{A} = \mathbf{C}.$$

Origin: S Ex. 3.23; H §16.5

Solution. Entrywise,

$$(\lambda_i - \lambda_j)x_{ij} = c_{ij}.$$

Thus a solution exists exactly when $c_{ii} = 0$ for every i . Under this condition,

$$x_{ij} = \frac{c_{ij}}{\lambda_i - \lambda_j} \quad (i \neq j), \quad x_{ii} \in \mathbb{R} \quad \text{arbitrary.}$$


Exercise 12.7 (A norm bound for a Sylvester solution). Let $\mathbf{A} \in \mathbb{R}^{m \times m}$, $\mathbf{B} \in \mathbb{R}^{n \times n}$ satisfy $\mathbf{A} \succ \mathbf{O}$, $\mathbf{B} \succ \mathbf{O}$, let $\mathbf{C} \in \mathbb{R}^{m \times n}$, and let $\mathbf{X}$ solve $\mathbf{AX} + \mathbf{XB} = \mathbf{C}$. Prove

$$\|\mathbf{X}\|_F \leq \frac{\|\mathbf{C}\|_F}{\lambda_{\min}(\mathbf{A}) + \lambda_{\min}(\mathbf{B})}.$$

Origin: Course formulation from S Ex. 3.23; H §16.5

Solution. Use Corollary 12.18. Orthogonal invariance of the Frobenius norm gives

$$\begin{aligned} \|\mathbf{X}\|_F^2 &= \sum_{i,j} \frac{y_{ij}^2}{(\lambda_i + \mu_j)^2} \\ &\leq \frac{\sum_{i,j} y_{ij}^2}{(\lambda_{\min}(\mathbf{A}) + \lambda_{\min}(\mathbf{B}))^2} = \frac{\|\mathbf{C}\|_F^2}{(\lambda_{\min}(\mathbf{A}) + \lambda_{\min}(\mathbf{B}))^2}. \end{aligned}$$


Exercise 12.8 (A matrix recurrence). Let $\mathbf{A} \in \mathbb{R}^{m \times m}$, $\mathbf{B} \in \mathbb{R}^{n \times n}$, and $\mathbf{X}_0 \in \mathbb{R}^{m \times n}$. If $\mathbf{X}_{k+1} = \mathbf{A}\mathbf{X}_k\mathbf{B}$ for $k \geq 0$, prove

$$\text{vec } \mathbf{X}_k = (\mathbf{B}' \otimes \mathbf{A})^k \text{vec } \mathbf{X}_0 = (\mathbf{B}'^k \otimes \mathbf{A}^k) \text{vec } \mathbf{X}_0$$

and

$$\|\mathbf{X}_k\|_F \leq \|\mathbf{A}\|_2^k \|\mathbf{B}\|_2^k \|\mathbf{X}_0\|_F.$$

Origin: A&M Ex. 10.18; H Eq. (16.1.22)

Solution. Roth's identity gives $\text{vec } \mathbf{X}_{k+1} = (\mathbf{B}' \otimes \mathbf{A}) \text{vec } \mathbf{X}_k$. Induction and the mixed-product rule give the two equalities. The vec isometry and induced-norm inequality give

$$\begin{aligned} \|\mathbf{X}_k\|_F &\leq \|\mathbf{B}'^k \otimes \mathbf{A}^k\|_2 \|\mathbf{X}_0\|_F \\ &= \|\mathbf{B}^k\|_2 \|\mathbf{A}^k\|_2 \|\mathbf{X}_0\|_F \leq \|\mathbf{B}\|_2^k \|\mathbf{A}\|_2^k \|\mathbf{X}_0\|_F. \end{aligned}$$



Exercise 12.9 (Frobenius adjoint of a matrix operator). Prove that the operator in Definition 12.15 has a unique Frobenius adjoint and that

$$\mathcal{L}^*(\mathbf{Y}) = \sum_{l=1}^k \mathbf{A}'_l \mathbf{Y} \mathbf{B}'_l, \quad \mathbf{M}_{\mathcal{L}^*} = \mathbf{M}'_{\mathcal{L}}.$$

Origin: Course formulation from A&M Ex. 10.20; H Thm. 16.2.2

Solution. Cyclicity of trace gives

$$\mathrm{tr}\{\mathbf{Y}' \mathbf{A}_l \mathbf{X} \mathbf{B}_l\} = \mathrm{tr}\{\mathbf{B}_l \mathbf{Y}' \mathbf{A}_l \mathbf{X}\} = \mathrm{tr}\{(\mathbf{A}'_l \mathbf{Y} \mathbf{B}'_l)' \mathbf{X}\}.$$

Thus the displayed map is a Frobenius adjoint. If $\mathcal{G}$ is another one, then, for every $\mathbf{X}$ and $\mathbf{Y}$,

$$\left\langle \mathbf{X}, \mathcal{G}(\mathbf{Y}) - \sum_{l=1}^k \mathbf{A}'_l \mathbf{Y} \mathbf{B}'_l \right\rangle_F = 0.$$

For fixed $\mathbf{Y}$, put

$$\Delta = \mathcal{G}(\mathbf{Y}) - \sum_l \mathbf{A}_l' \mathbf{Y} \mathbf{B}_l'.$$

Setting $\mathbf{X} = \Delta$ gives

$$0 = \langle \Delta, \Delta \rangle_F = \|\Delta\|_F^2 \implies \Delta = \mathbf{O}.$$

Thus $\mathcal{G} = \mathcal{L}^*$. The coefficient of the l th term is

$$\mathbf{B}_l \otimes \mathbf{A}_l' = (\mathbf{B}_l' \otimes \mathbf{A}_l)',$$

and summation gives $\mathbf{M}_{\mathcal{L}^*} = \mathbf{M}_{\mathcal{L}}'$. ■

Exercise 12.10 (Minimum-norm matrix least squares). Let $\mathbf{A} \in \mathbb{R}^{r \times m}$, $\mathbf{B} \in \mathbb{R}^{n \times s}$, and $\mathbf{C} \in \mathbb{R}^{r \times s}$. Prove

$$(\mathbf{A} \otimes \mathbf{B})^+ = \mathbf{A}^+ \otimes \mathbf{B}^+.$$

Then prove that the unique minimum-Frobenius-norm matrix among the minimizers of $\|\mathbf{A}\mathbf{X}\mathbf{B} -$

$\mathbf{C}\|_F$ is

$$\widehat{\mathbf{X}} = \mathbf{A}^+ \mathbf{C} \mathbf{B}^+.$$

Origin: A&M Exs. 10.25(c), 10.49; H Eqs. (16.1.24), (16.5.3)

Solution. Apply the construction in the proof of Theorem 12.4 to SVDs of $\mathbf{A}$ and $\mathbf{B}$; reciprocating the positive diagonal entries gives

$$(\mathbf{A} \otimes \mathbf{B})^+ = \mathbf{A}^+ \otimes \mathbf{B}^+.$$

The coefficient of $\mathbf{A} \mathbf{X} \mathbf{B}$ is $\mathbf{B}' \otimes \mathbf{A}$. Hence if $\mathbf{B} = \mathbf{U} \mathbf{D} \mathbf{V}'$ is an SVD, then $\mathbf{B}' = \mathbf{V} \mathbf{D} \mathbf{U}'$ and $(\mathbf{B}')^+ = \mathbf{U} \mathbf{D}^+ \mathbf{V}' = (\mathbf{B}^+)'$. Therefore

$$\begin{aligned} \text{vec } \widehat{\mathbf{X}} &= (\mathbf{B}' \otimes \mathbf{A})^+ \text{vec } \mathbf{C} \\ &= ((\mathbf{B}^+)' \otimes \mathbf{A}^+) \text{vec } \mathbf{C} = \text{vec}(\mathbf{A}^+ \mathbf{C} \mathbf{B}^+). \end{aligned}$$

The Moore–Penrose least-squares theorem and the vec isometry give both minimum properties. ■

Exercise 12.11 (A transposed matrix equation). For $\mathbf{X}, \mathbf{C} \in \mathbb{R}^{n \times n}$, characterize all solutions of $\mathbf{X} + \mathbf{X}' = \mathbf{C}$. If the equation is consistent, identify its minimum-Frobenius-norm solution and prove

$$(\mathbf{I}_{n^2} + \mathbf{K}_{nn})^+ = \frac{1}{2}\mathbf{N}_n.$$

Origin: A&M Exs. 11.14, 11.17; H Eqs. (16.5.4)–(16.5.6)

Solution. The left side is symmetric, so consistency requires $\mathbf{C} = \mathbf{C}'$. Under this condition,

$$\mathbf{X} + \mathbf{X}' = \mathbf{C} \iff \mathbf{X} = \frac{1}{2}\mathbf{C} + \mathbf{S}, \quad \mathbf{S}' = -\mathbf{S}.$$

Theorem 12.25 and the vec isometry make the symmetric and skew-symmetric parts orthogonal in the Frobenius inner product; hence the norm is minimized at $\mathbf{S} = \mathbf{O}$. Since $\mathbf{I} + \mathbf{K} = 2\mathbf{N}$ and $\mathbf{N}' = \mathbf{N}^2 = \mathbf{N}$, the four Penrose equations give $(2\mathbf{N})^+ = \mathbf{N}/2$. ■

Exercise 12.12 (Nearest patterned parts). For $\mathbf{X} \in \mathbb{R}^{n \times n}$, set

$$\mathbf{S}_0 = \frac{\mathbf{X} + \mathbf{X}'}{2}, \quad \mathbf{T}_0 = \frac{\mathbf{X} - \mathbf{X}'}{2}.$$

Prove

$$\begin{aligned} \|\mathbf{X} - \mathbf{S}\|_F^2 &= \|\mathbf{T}_0\|_F^2 + \|\mathbf{S} - \mathbf{S}_0\|_F^2 & (\mathbf{S} = \mathbf{S}'), \\ \|\mathbf{X} - \mathbf{T}\|_F^2 &= \|\mathbf{S}_0\|_F^2 + \|\mathbf{T} - \mathbf{T}_0\|_F^2 & (\mathbf{T}' = -\mathbf{T}). \end{aligned}$$

Deduce that $\mathbf{S}_0$ and $\mathbf{T}_0$ are the respective unique minimizers.

Origin: A&M Exs. 11.14, 11.17; H Eq. (16.4.19)

Solution. $\mathbf{S}_0$ and $\mathbf{S} - \mathbf{S}_0$ are symmetric; $\mathbf{T}_0$ is skew-symmetric. Theorem 12.25 and the vec isometry give

$$\langle \mathbf{T}_0, \mathbf{S} - \mathbf{S}_0 \rangle_F = 0, \quad \mathbf{X} - \mathbf{S} = \mathbf{T}_0 - (\mathbf{S} - \mathbf{S}_0),$$

which gives the first identity. Similarly, $\mathbf{T} - \mathbf{T}_0$ is skew-symmetric and

$$\langle \mathbf{S}_0, \mathbf{T} - \mathbf{T}_0 \rangle_F = 0, \quad \mathbf{X} - \mathbf{T} = \mathbf{S}_0 - (\mathbf{T} - \mathbf{T}_0),$$

which gives the second. Each remainder norm vanishes at exactly one feasible matrix. ■

Exercise 12.13 (Weights in symmetric coordinates). For symmetric $\mathbf{S} = (s_{ij})$, put $\mathbf{h} = \text{vech } \mathbf{S}$. Prove

$$\|\mathbf{S}\|_F^2 = \mathbf{h}' \mathbf{D}'_n \mathbf{D}_n \mathbf{h} = \sum_{i=1}^n s_{ii}^2 + 2 \sum_{i>j} s_{ij}^2.$$

For arbitrary $\mathbf{X}$, prove that $\mathbf{D}_n^+ \text{vec } \mathbf{X}$ retains diagonal entries and replaces each lower-triangular entry by $(x_{ij} + x_{ji})/2$.

Origin: A&M Ex. 11.30; H §16.4b and Ex. 16.27

Solution. Since $\text{vec } \mathbf{S} = \mathbf{D}_n \mathbf{h}$,

$$\|\mathbf{S}\|_F^2 = (\mathbf{D}_n \mathbf{h})' \mathbf{D}_n \mathbf{h} = \mathbf{h}' \mathbf{D}'_n \mathbf{D}_n \mathbf{h}.$$

The diagonal weights of $\mathbf{D}'_n \mathbf{D}_n$ are 1 for diagonal coordinates and 2 for off-diagonal coordinates. Also,

$$\mathbf{D}_n^+ \text{vec } \mathbf{X} = \mathbf{L}_n \mathbf{N}_n \text{vec } \mathbf{X} = \text{vech } \frac{\mathbf{X} + \mathbf{X}'}{2},$$

which gives the entry formula. ■

Exercise 12.14 (Orthogonal congruence in vech coordinates). Let $\mathbf{Q} \in \mathbb{R}^{n \times n}$ be orthogonal, $\mathbf{G}_n = \mathbf{D}_n' \mathbf{D}_n$, and $\mathbf{C}_\mathbf{Q} = \mathbf{D}_n^+ (\mathbf{Q} \otimes \mathbf{Q}) \mathbf{D}_n$. Prove

$$\mathbf{C}_\mathbf{Q}' \mathbf{G}_n \mathbf{C}_\mathbf{Q} = \mathbf{G}_n.$$

Thus $\mathbf{C}_\mathbf{Q}$ preserves the $\mathbf{G}_n$ -weighted inner product even when it does not preserve the ordinary Euclidean inner product on vech coordinates.

Origin: Course formulation from A&M Ex. 11.31; H Eq. (16.4.25)

Solution. For arbitrary $\mathbf{h} = \text{vech } \mathbf{S}$ with $\mathbf{S} = \mathbf{S}'$, duplication and orthogonal invariance give

$$\mathbf{h}' \mathbf{C}_\mathbf{Q}' \mathbf{G}_n \mathbf{C}_\mathbf{Q} \mathbf{h} = \|\mathbf{D}_n \mathbf{C}_\mathbf{Q} \mathbf{h}\|_2^2 = \|\mathbf{Q} \mathbf{S} \mathbf{Q}'\|_F^2 = \|\mathbf{S}\|_F^2 = \mathbf{h}' \mathbf{G}_n \mathbf{h}.$$

Both coefficient matrices are symmetric. Equality of their quadratic forms for every $\mathbf{h}$ gives the matrix identity. ■

Exercise 12.15 (A symmetric matrix basis equation). Let $\mathbf{A}_1, \dots, \mathbf{A}_k, \mathbf{C}$ be symmetric $n \times n$ matrices and put

$$\mathbf{G} = (\text{vech } \mathbf{A}_1, \dots, \text{vech } \mathbf{A}_k), \quad \mathbf{H} = (\text{vec } \mathbf{A}_1, \dots, \text{vec } \mathbf{A}_k).$$

Prove

$$\sum_{i=1}^k \theta_i \mathbf{A}_i = \mathbf{C} \iff \mathbf{G}\boldsymbol{\theta} = \text{vech } \mathbf{C}, \quad \mathbf{H} = \mathbf{D}_n \mathbf{G}, \quad \text{rank } \mathbf{H} = \text{rank } \mathbf{G}.$$

Derive the normal equations for minimizing $\|\sum_i \theta_i \mathbf{A}_i - \mathbf{C}\|_F^2$ in vech coordinates.

Origin: H Ex. 16.30

Solution. Linearity and injectivity of vech on symmetric matrices give the first equivalence. Columnwise duplication gives $\mathbf{H} = \mathbf{D}_n \mathbf{G}$. Since $\mathbf{D}_n$ has full column rank,

$$\text{null}(\mathbf{D}_n \mathbf{G}) = \text{null } \mathbf{G}, \quad \text{rank } \mathbf{H} = \text{rank } \mathbf{G}.$$

Since $\text{vec } \mathbf{A}_i = \mathbf{D}_n \text{vech } \mathbf{A}_i$ and $\text{vec } \mathbf{C} = \mathbf{D}_n \text{vech } \mathbf{C}$, the objective is

$$(\mathbf{G}\boldsymbol{\theta} - \text{vech } \mathbf{C})' \mathbf{D}_n' \mathbf{D}_n (\mathbf{G}\boldsymbol{\theta} - \text{vech } \mathbf{C}).$$

Its normal equations are

$$\mathbf{G}'\mathbf{D}'_n\mathbf{D}_n\mathbf{G}\hat{\boldsymbol{\theta}} = \mathbf{G}'\mathbf{D}'_n\mathbf{D}_n \text{vech } \mathbf{C}.$$

